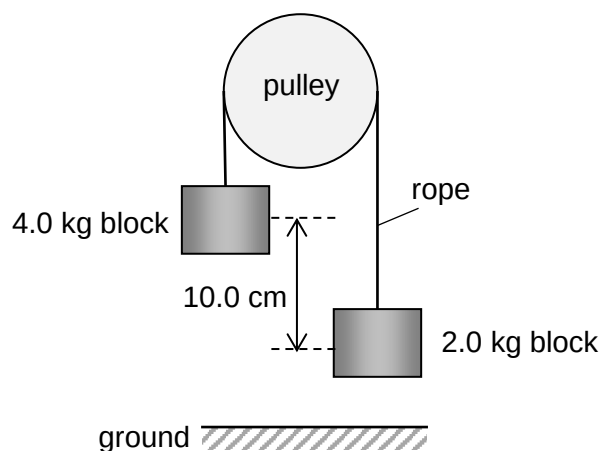


- 12 The figure below shows a light inextensible rope that passes over a light smooth pulley with two blocks of masses 4.0 kg and 2.0 kg attached to its two ends. The two blocks are initially at rest with the 4.0 kg block 10.0 cm above the 2.0 kg block.



What is the speed of the blocks when they are at the same height above the ground?

- A** 0.57 m s⁻¹ **B** 0.81 m s⁻¹ **C** 0.99 m s⁻¹ **D** 1.4 m s⁻¹

Ans: A

$$\text{Loss in GPE} = 4.0 \times 9.8 \times (5.0 \times 10^{-2}) - 2.0 \times 9.8 \times (5.0 \times 10^{-2}) = 0.98 \text{ J}$$

$$\text{Gain in KE} = \frac{1}{2} \times (2.0 + 4.0) \times v^2 = 0.98 \text{ J}$$

$$\rightarrow v = 0.57 \text{ m s}^{-1}$$

12 A red DD of length l is turned about the point D through an angle θ